Vehicle Parameter Measurement Test Procedure

F1/10th — Traxxas Slash 4×4 Chassis

Bachelor Project

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Contents

1 Introduction

This document defines the full test procedure for physically measuring every relevant parameter of the F1/10th platform. All measurements are performed on the bare **Traxxas Slash 4×4** chassis equipped with *only* the battery and motor (Traxxas 3351R). The ESC and RC receiver have been removed and are *not* present during these measurements.

The measured values feed directly into:

- The MPC vehicle model (`mpc_types.h`, `vehicle_model.h`)
- The planning pipeline (`vehicle_params.yaml`)
- The ROS 2 control stack (`vesc.yaml`, EKF parameters)

2 Required Equipment

- Digital kitchen/postal scale — resolution ≤ 1 g (range ≥ 5 kg)
- Vernier caliper or digital caliper — resolution 0.01 mm
- Steel ruler or measuring tape — 1 m, 1 mm graduation
- Set square / right-angle tool
- Flat, level surface (table)
- Pencil / marker
- Masking tape
- Spirit level (optional, for verifying surface is flat)
- 4 small identical blocks or coins (for lifting individual wheels)

Note: The vehicle is weighed with the Traxxas 4×4 Slash chassis containing **only** the battery and motor. No ESC, no receiver, no LiDAR, no compute stack. Once these components are added later, the measurements in Section 4 should be repeated.

3 Vehicle Diagram and Nomenclature

The following conventions are used throughout.

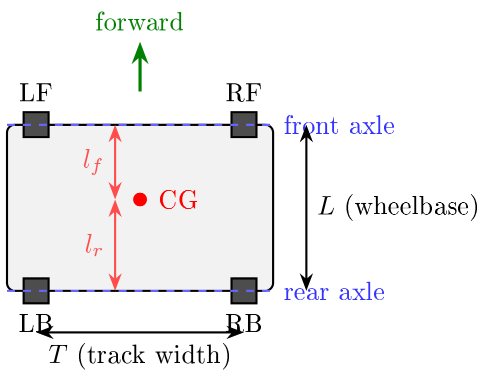

Figure 1: Top-down schematic with parameter naming conventions.

Table 1: Parameter symbols and descriptions.

Symbol	Parameter	Unit
m	Total vehicle mass	kg
m_{LF}	Mass on left-front wheel	kg
m_{RF}	Mass on right-front wheel	kg
m_{LB}	Mass on left-rear wheel	kg
m_{RB}	Mass on right-rear wheel	kg
L	Wheelbase (front-to-rear axle)	m
l_f	Distance from CG to front axle	m
l_r	Distance from CG to rear axle	m
T_f	Front track width (centre-to-centre)	m
T_r	Rear track width (centre-to-centre)	m
d_w	Wheel (tire) outer diameter	m
r_w	Wheel (tire) rolling radius	m
w_t	Tire width	m
h_{cg}	Centre-of-gravity height	m
L_{total}	Overall vehicle length	m
W_{total}	Overall vehicle width	m
I_z	Yaw moment of inertia	$\mathrm{kg\,m^2}$

4 Test 1: Mass and Weight Distribution

4.1 Purpose

Determine the total mass m and the individual corner weights m_{LF}, m_{RF}, m_{LB}, m_{RB}. From these, the longitudinal CG position (l_f, l_r) and the lateral CG offset are computed.

4.2 Setup

1. Place the scale on a flat, level surface.
2. Set wheels straight ahead (zero steering angle).
3. The vehicle contains **only** the battery and motor — verify that no ESC or receiver is installed.

4.3 Procedure

4.3.1 Individual corner weights

The total mass is derived from the sum of all four corner weights, so no separate whole-vehicle weighing step is needed. (An independent total-mass measurement can be added later as a verification.)
For each corner (LF, RF, LB, RB):

1. Place *one* wheel on the scale.
2. Support the other three wheels at the *same* height as the scale surface (use spacer blocks of identical thickness so the chassis is level).
3. Record the reading as m_{XX}.
4. Repeat for all four corners.

Note: When an independent total-mass measurement becomes available, verify that $m_{\mathrm{LF}} + m_{\mathrm{RF}} + m_{\mathrm{LB}} + m_{\mathrm{RB}} \approx m$. Any deviation $> 2\,\%$ indicates the chassis was not level during corner weighing.

4.4 Data Recording

Quantity	**Measured value**	
m_{LF}	0.572 kg	
m_{RF}	0.501 kg	
m_{LB}	0.604 kg	
m_{RB}	0.616 kg	
$m = \sum m_i$	2.293 kg	(computed)

4.5 Derived quantities

Front axle mass and rear axle mass:

$$m_f = m_{\mathrm{LF}} + m_{\mathrm{RF}}, \qquad m_r = m_{\mathrm{LB}} + m_{\mathrm{RB}} \tag{1}$$

Longitudinal CG position (requires wheelbase L from Test 2):

$$l_r = \frac{m_f}{m} L, \qquad l_f = L - l_r \tag{2}$$

Lateral CG offset from centreline:

$$\Delta y_{\mathrm{cg}} = \frac{(m_{\mathrm{RF}} + m_{\mathrm{RB}}) - (m_{\mathrm{LF}} + m_{\mathrm{LB}})}{m} \cdot \frac{T}{2} \tag{3}$$

where T is the average track width. A positive value means the CG is offset to the right.

5 Test 2: Wheelbase

5.1 Purpose

Measure the wheelbase L, the distance between the front and rear axle centres.

5.2 Procedure

1. Place the vehicle on a flat surface with wheels straight ahead.
2. Mark the **centre of the front axle** on the ground (midpoint between the two front wheel hubs, projected vertically downward; a plumb line or set square helps).
3. Mark the **centre of the rear axle** in the same manner.
4. Measure the distance between the two marks with a caliper or ruler.
5. Repeat 3 times and take the average.

Alternative (direct hub-to-hub): Measure from the centre of a front wheel hub to the centre of the corresponding rear wheel hub on the same side. Repeat on both sides; they should agree within 1 mm.

5.3 Data Recording

Measurement	Value
L	0.326 m

Now compute l_f and l_r using Eq. (2) and the masses from Test 1.

l_f	0.1734 m
l_r	0.1526 m

6 Test 3: Track Width (Front and Rear)

6.1 Purpose

Measure the distance between the left and right wheel contact-patch centres on each axle.

6.2 Procedure

1. Set wheels straight ahead.
2. **Front track T_f:** Measure the distance between the centres of the two front tyres at ground level (contact-patch centre to contact-patch centre). Alternatively, measure hub-to-hub if the wheels are symmetric.
3. **Rear track T_r:** Same for the rear axle.
4. Repeat 3 times each.

6.3 Data Recording

Measurement	Value
T_f	0.231 m
T_r	0.231 m

7 Test 4: Wheel and Tyre Dimensions

7.1 Purpose

Measure wheel diameter d_w, rolling radius r_w, and tyre width w_t.

7.2 Procedure

7.2.1 Wheel diameter

1. Remove one wheel from the vehicle.
2. Using a caliper, measure the outer tyre diameter at three orientations (rotate 60° between measurements) to account for out-of-round.
3. Record the average as d_w.

7.2.2 Rolling radius (loaded)

1. Place the vehicle on a flat surface (all four wheels on the ground, full weight applied).
2. Measure the height from the ground to the centre of a wheel hub (use a caliper or ruler against the axle).
3. This is the **loaded rolling radius** r_w.
4. Repeat for all four wheels and average.

Note: The loaded rolling radius is generally slightly smaller than $d_w/2$ due to tyre deformation under load. The rolling radius is the value needed for ERPM-to-speed conversion.

7.2.3 Tyre width

1. With the wheel removed, measure the tread width (ground contact width) with a caliper.
2. Record as w_t.

7.3 Data Recording

Measurement	Value
d_w	0.100 m
r_w (loaded, LF)	0.0500 m
r_w (loaded, RF)	0.0505 m
r_w (loaded, LB)	0.0505 m
r_w (loaded, RB)	0.0505 m
r_w (average)	0.050 38 m
w_t	0.043 m

8 Test 5: Overall Dimensions

8.1 Purpose

Measure the overall vehicle length and width for collision avoidance margins.

8.2 Procedure

1. **Overall length L_{total}:** Measure from the furthest-forward point (front bumper) to the furthest-rearward point (rear bumper), parallel to the longitudinal axis.
2. **Overall width W_{total}:** Measure the widest point of the vehicle (typically the outer edges of the tyres).

8.3 Data Recording

Measurement	Value
L_{total}	0.510 m
W_{total}	0.271 m

9 Test 6: Centre-of-Gravity Height

9.1 Purpose

Estimate the CG height h_{cg} above the ground. This is needed for load-transfer calculations and rollover analysis.

9.2 Method — Tilt Test

1. Place the vehicle on a flat, level surface.
2. Measure and record the rear-axle weight $m_r^{\text{level}} = m_{\text{LB}} + m_{\text{RB}}$ (from Test 1).
3. Raise the **front axle** by a known height Δh using a block or stack of known thickness. Keep the rear wheels on the scale.
4. Record the new rear-axle scale reading m_r^{tilted}.
5. Compute the tilt angle:
$$\theta = \arcsin\left(\frac{\Delta h}{L}\right) \tag{4}$$
6. Compute CG height:
$$h_{\text{cg}} = \frac{(m_r^{\text{tilted}} - m_r^{\text{level}})\,L}{m\,\tan\theta} \;+\; r_w \tag{5}$$
(The $+\,r_w$ accounts for the fact that the axle is one wheel radius above the ground.)

Note: Use a tilt of roughly 5° to 15° for best accuracy. Too small and the weight shift is in the noise; too large and the battery may shift.

9.3 Data Recording

Quantity	Value	
Δh (front raised)	____________	m
θ	____________	°
m_r^{level}	____________	kg
m_r^{tilted}	____________	kg
h_{cg} (computed)	____________	m

10 Test 7: Yaw Moment of Inertia (Estimation)

10.1 Purpose

Estimate the yaw moment of inertia I_z about the vertical axis through the CG.

10.2 Method A — Bifilar (Two-String) Pendulum

This is the most practical method for a small vehicle.

1. Suspend the vehicle from two parallel strings of equal length l_s, attached at two points on the chassis separated by a horizontal distance $2d$ (symmetric about the CG).
2. Allow the vehicle to oscillate in yaw (twist about the vertical axis) with a small amplitude ($< 10°$).

3. Time $N \geq 10$ oscillation periods and compute the average period T_{osc}.

4. Calculate:

$$I_z = \frac{m\,g\,d^2\,T_{\text{osc}}^2}{4\,\pi^2\,l_s} \tag{6}$$

10.3 Method B — Compound Pendulum (simpler)

1. Suspend the vehicle from a single pivot point (e.g. a bolt through a convenient hole) at a known distance R from the CG.

2. Let it swing as a compound pendulum in the yaw plane and measure the period T_{osc}.

3. Calculate:

$$I_z = \frac{m\,g\,R\,T_{\text{osc}}^2}{4\,\pi^2} - m\,R^2 \tag{7}$$

10.4 Data Recording

Quantity	Value	
Method used	____________	
l_s or R	____________	m
d (bifilar)	____________	m
Oscillations N	____________	
Total time	____________	s
T_{osc}	____________	s
I_z (computed)	____________	kg m^2

11 Test 8: Steering Geometry

11.1 Purpose

Measure the maximum mechanical steering angle and characterise the Ackermann steering geometry.

11.2 Procedure

11.2.1 Maximum steering angle

1. With the servo/steering linkage connected (but no ESC powered), manually turn the front wheels to full lock in each direction.

2. Using a protractor or by measuring geometry (see below), determine the maximum steering angle $\delta_{\max}$ for both left and right.

Geometric method for steering angle:

1. Set wheels to full lock.

2. Mark a straight reference line along the chassis centreline on the ground.

3. Mark the line along the wheel's longitudinal axis (use a ruler against the tyre sidewall).

4. The angle between these two lines is δ.

11.2.2 Ackermann check

1. At full lock, measure the inner wheel angle δ_{in} and the outer wheel angle δ_{out}.
2. For perfect Ackermann geometry:

$$\cot\delta_{\text{out}} - \cot\delta_{\text{in}} = \frac{T_f}{L} \tag{8}$$

3. Record both angles to characterise the actual Ackermann percentage.

11.3 Data Recording

Measurement	Value
$\delta_{\text{max,left}}$	________ ° (________ rad)
$\delta_{\text{max,right}}$	________ ° (________ rad)
δ_{in} (full lock)	________ °
δ_{out} (full lock)	________ °

12 Summary of Results

Transfer all final measured and computed values here.

Table 2: Complete vehicle parameter summary (Traxxas Slash 4×4, battery + motor only, no ESC/receiver).

Parameter	Symbol	Value	Config field
Total mass	m	2.293 kg	`mass`
Front-left corner mass	m_{LF}	0.572 kg	—
Front-right corner mass	m_{RF}	0.501 kg	—
Rear-left corner mass	m_{LB}	0.604 kg	—
Rear-right corner mass	m_{RB}	0.616 kg	—
Wheelbase	L	0.326 m	`wheelbase`
CG to front axle	l_f	0.1734 m	`lf`
CG to rear axle	l_r	0.1526 m	`lr`
Front track width	T_f	0.231 m	—
Rear track width	T_r	0.231 m	—
Wheel diameter	d_w	0.100 m	—
Rolling radius	r_w	0.050 38 m	—
Tyre width	w_t	0.043 m	—
Overall length	L_{total}	0.510 m	`length`
Overall width	W_{total}	0.271 m	`width`
CG height	h_{cg}	________ m	`cg_height`
Yaw moment of inertia	I_z	________ kg m^2	`inertia`
Max steering angle (left)	$\delta_{\text{max,L}}$	________ rad	`max_steering_angle`
Max steering angle (right)	$\delta_{\text{max,R}}$	________ rad	—

13 Mapping to Configuration Files

After completing all measurements, update the following files with the measured values:

13.1 `vehicle_params.yaml` (f1tenth_planning/config/)

```
vehicle:
  wheelbase:        <L>
  width:            <W_total>
advanced:
  lf:               <l_f>
  lr:               <l_r>
  length:           <L_total>
  mass:             <m>
  inertia:          <I_z>
  cg_height:        <h_cg>
  max_steering_angle: <delta_max>
```

13.2 `mpc_types.h` (MPC/include/)

```
wheelbase_meters             = FP_FROM_FLOAT(<L>)
maximum_steering_angle_radians = FP_FROM_FLOAT(<delta_max>)
```

13.3 `common.py` (f1tenth_parameters/)

```
DEFAULT_WHEELBASE = <L>
```

14 Notes and Considerations

- **Current configuration:** The vehicle is the Traxxas Slash 4×4 chassis with *only* the battery (3S LiPo) and the Traxxas 3351R motor installed. There is **no ESC** and **no RC receiver** on the vehicle during these measurements.
- **Re-measurement after assembly:** Once the VESC, Jetson compute board, LiDAR, and any other components are added, the mass-dependent parameters (m, l_f, l_r, h_{cg}, I_z) **must** be re-measured, as the added mass and its placement will shift the CG and change the inertia.
- **Geometric parameters are stable:** Wheelbase L, track widths T_f, T_r, wheel diameter d_w, and overall dimensions do not change when electronics are added — measure them once.
- **Tyre inflation:** These are foam-filled RC tyres, so inflation is not a variable; however, tyre wear will reduce d_w and r_w over time.
- **Units:** All values should be recorded in SI base units (metres, kilograms, radians, seconds) for direct use in the control software.